

Department of Computer Applications

II MCA – Java Programming Lab – Handson Worksheet 3

Reference Variables, Strings & Arrays

Assignment 1: Library Login Story

A college digital library allows students to log in. Two users appear to type the same username, but the system must verify whether both references actually point to the same object or only contain the same text.

Problem Statement:

Compare two strings created using literal and new String(). Demonstrate == and equals().

Sample Input/Output:

Output:

s1 == s2 : false

s1.equals(s2) : true

Skeleton Code Template:

```
class LoginCheck {  
    public static void main(String[] args) {  
        String s1 = "user";  
        String s2 = new String("user");  
  
        // comparison logic  
    }  
}
```

Viva Questions:

1. What is string pooling?
2. Difference between == and equals()?
3. Why are strings immutable?

Assignment 2: Chat Message Cleaner

Students send messages with extra spaces and unwanted words. The system must clean and standardize messages.

Problem Statement:

Use trim(), toLowerCase(), and replace() to clean the message.

Sample Input/Output:

Input: Hello BAD World

Output: hello good world

Skeleton Code Template:

```
import java.util.*;
class Cleaner {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String msg = sc.nextLine();

        // cleaning logic
    }
}
```

Viva Questions:

1. What does trim() do?
2. What is immutability?
3. Does replace() modify original string?

Assignment 3: Password Strength Checker

Users must create strong passwords for the portal.

Problem Statement:

Check length, digits, and uppercase letters using string traversal.

Sample Input/Output:

Input: Pass123

Output: Strong password

Skeleton Code Template:

```
import java.util.*;
class PasswordCheck {
    public static void main(String[] args) {
        String pwd = new Scanner(System.in).nextLine();

        // validation
    }
}
```

Viva Questions:

1. How to access characters?
2. What is charAt()?
3. Why loops needed?

Assignment 4: Student Marks Analyzer

Teacher stores marks digitally and needs quick statistics.

Problem Statement:

Store marks in array and compute average, min, max.

Sample Input/Output:

Input: 5 marks

Output: Avg=70 Max=90 Min=50

Skeleton Code Template:

```
class Marks {  
    public static void main(String[] args) {  
        int[] marks = new int[5];  
  
        // calculations  
    }  
}
```

Viva Questions:

1. Define array.
2. Default values of int array?
3. How to traverse?

Assignment 5: Word Counter

News app counts words in articles.

Problem Statement:

Split sentence into words using `split()` and count.

Sample Input/Output:

Input: Java is fun

Output: 3 words

Skeleton Code Template:

```
import java.util.*;  
class WordCount {  
    public static void main(String[] args) {  
        String line = new Scanner(System.in).nextLine();  
  
        // split and count  
    }  
}
```

Viva Questions:

1. split() returns what?
2. What is regex?
3. How to get array size?

Assignment 6: Email Validator

Registration system must validate emails.

Problem Statement:

Check '@' and '.' positions using indexOf().

Sample Input/Output:

Input: test@gmail.com

Output: Valid

Skeleton Code Template:

```
import java.util.*;
class EmailCheck {
    public static void main(String[] args) {
        String email = new Scanner(System.in).nextLine();

        // validation
    }
}
```

Viva Questions:

1. Use of indexOf()?
2. What is substring()?
3. Simple validation rule?

Assignment 7: Playlist Manager

A music app stores songs and lets users search.

Problem Statement:

Search song name in string array and print position.

Sample Input/Output:

Input: C

Output: Found at position 3

Skeleton Code Template:

```
class Playlist {
    public static void main(String[] args) {
        String[] songs = {"A","B","C"};
```

```
        // search logic
    }
}
```

Viva Questions:

1. What is linear search?
2. equals vs ==?
3. Time complexity?

Assignment 8: Text Analyzer

Analytics tool counts characters in reviews.

Problem Statement:

Count vowels, digits, spaces using charAt().

Sample Input/Output:

Input: Hello 123

Output: Vowels=2 Digits=3 Spaces=1

Skeleton Code Template:

```
import java.util.*;
class Analyzer {
    public static void main(String[] args) {
        String text = new Scanner(System.in).nextLine();

        // counting
    }
}
```

Viva Questions:

1. What does charAt() return?
2. Character methods?
3. Logic for counting?

Assignment 9: Attendance Tracker

Teacher marks attendance digitally.

Problem Statement:

Use boolean array to store present/absent and count.

Sample Input/Output:

Output: Present=25 Absent=5

Skeleton Code Template:

```

class Attendance {
    public static void main(String[] args) {
        boolean[] att = new boolean[30];

        // count
    }
}

```

Viva Questions:

1. Default boolean value?
2. Why boolean type?
3. Counting method?

Assignment 10: Duplicate Name Finder

School system detects duplicate registrations.

Problem Statement:

Find duplicates in string array using equals().

Sample Input/Output:

Output: Duplicate: Ravi

Skeleton Code Template:

```

class DuplicateNames {
    public static void main(String[] args) {
        String[] names = {"A","B","A"};

        // nested loops
    }
}

```

Viva Questions:

1. Why equals()?
2. Nested loop complexity?
3. Better approaches?

Assignment 11: Story Editor

User edits large text repeatedly. Efficiency matters.

Problem Statement:

Append text using StringBuilder and display result.

Sample Input/Output:

Output: final paragraph displayed

Skeleton Code Template:

```
class Editor {  
    public static void main(String[] args) {  
        StringBuilder sb = new StringBuilder();  
  
        // append  
    }  
}
```

Viva Questions:

1. Why StringBuilder faster?
2. Mutable vs immutable?
3. Common methods?

Assignment 12: Customer Name Formatter

A CRM system stores customer names in inconsistent formats. The system must standardize them before saving.

Problem Statement:

Accept a full name. Convert to uppercase, remove extra spaces, and print initials.

Sample Input/Output:

Input: raja kumar

Output: RAJA KUMAR | Initials: R.K

Skeleton Code Template:

```
import java.util.*;  
class NameFormatter {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        String name = sc.nextLine();  
  
        // formatting logic  
    }  
}
```

Viva Questions:

1. What does toUpperCase() do?
2. How to split words?
3. Why strings are immutable?

Assignment 13: String Pool Experiment

A developer wants to understand how Java saves memory using the string pool.

Problem Statement:

Create multiple strings using literals and new. Compare references and count which share same memory.

Sample Input/Output:

Output:

s1 == s2 : true

s1 == s3 : false

Skeleton Code Template:

```
class StringPoolDemo {
    public static void main(String[] args) {
        String s1 = "java";
        String s2 = "java";
        String s3 = new String("java");

        // comparisons
    }
}
```

Viva Questions:

1. What is string constant pool?
2. Why literals share memory?
3. Difference between heap and pool?

Assignment 14: Bus Seat Reservation System

A bus has 20 seats. The system tracks booked seats and prints available ones.

Problem Statement:

Use an integer array to mark seats (0 free, 1 booked). Allow booking and display free seats.

Sample Input/Output:

Input: Book seat 5

Output: Seat 5 booked

Available seats: ...

Skeleton Code Template:

```
class BusSeats {
    public static void main(String[] args) {
        int[] seats = new int[20];

        // booking logic
    }
}
```


Viva Questions:

1. Default value of int array?
2. How to mark booking?
3. How to traverse array?

Assignment 15: Sentence Reverser

Digital Story:

A social media app displays captions in reverse order for a fun effect.

Problem Statement:

Accept a sentence, split into words, store in array, and print in reverse order.

Sample Input/Output:

Input: Java is powerful

Output: powerful is Java

Skeleton Code Template:

```
import java.util.*;
class ReverseSentence {
    public static void main(String[] args) {
        String line = new Scanner(System.in).nextLine();

        // reverse logic
    }
}
```

Viva Questions:

1. What does split() return?
2. How to reverse an array?
3. Difference between String and StringBuilder reverse()?